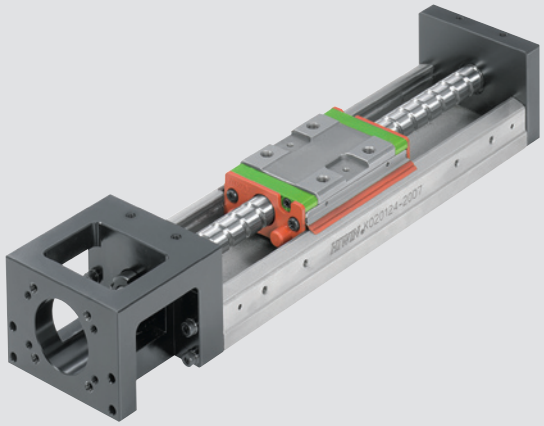
CRD Devices Ltd All Saints Industrial Estate, Shildon, Co, Durham, DL4 2RD
Tel: +44 (0)1388 778400 Fax: +44 (0)1388 778800
E: sales@crd-devices.co.uk W: www.crd-devices.co.uk

# Positioning Systems
## Linear Module



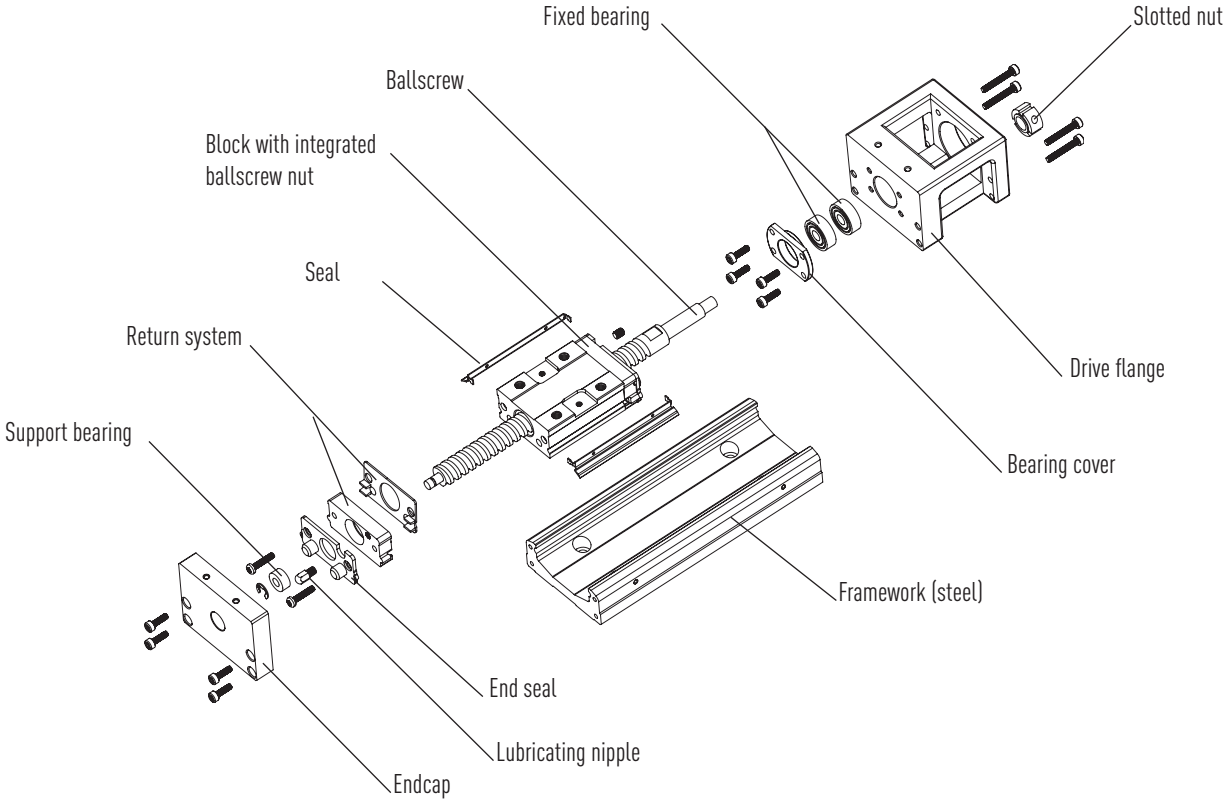
## 6. Linear Stages

### 6.1 Product Overview

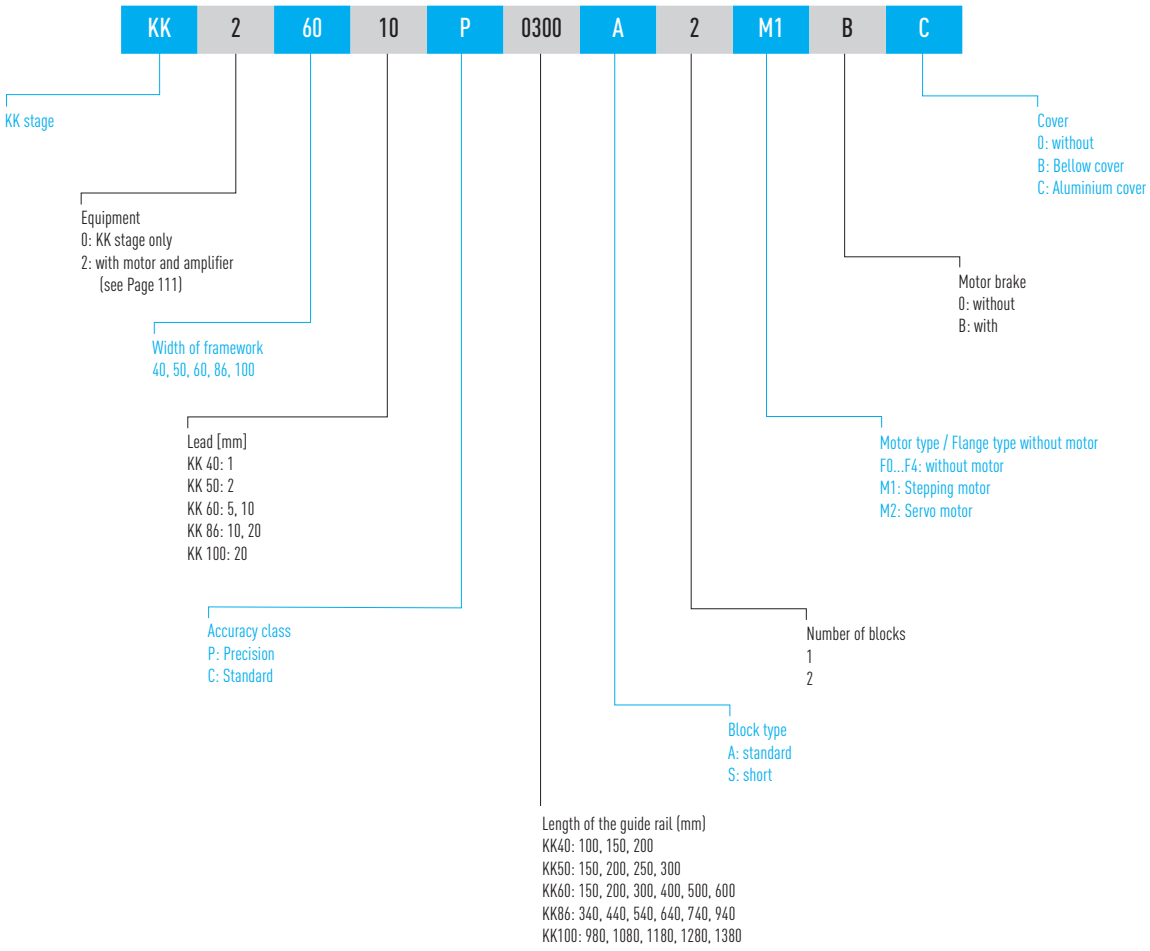
**Linear Stages with Ballscrew (KK Stages)**
HIWIN linear stages (KK stages) are compact positioning stages. The advance is generated by a ballscrew, which is mounted in a drive flange ready
to use by the motor. Movement is guided by a linear guideway. Various equipment versions and sizes adapt the linear stages to very different tasks and industries.

- Module for positioning tasks – KK linear stages with ballscrews from HIWIN can be used universally and are suitable as ready-to-mount stages for many different positioning tasks
- Lean and light – thanks to their compact and lean construction as well as light mass, KK stages can also be integrated into applications with little space.
- Flexible and adaptable – various servo motors, controllers, special models and accessories make KK stages suitable for universal use. KK stages can be supplied with or without a motor on request.
- Modular and multi-dimensional – multiple axis systems can be achieved easily with the KK stages.
- Adaptable and sturdy – KK stages can be equipped with a bellow cover or aluminum cover depending on the ambient requirements.
- Vacuum model possible
- Framework and block made of steel with surface corrosion protection
- Low maintenance

#### 6.1.1 Exploded View of the Linear Stages



## 6.1.2 Model Numbers for Linear Stages



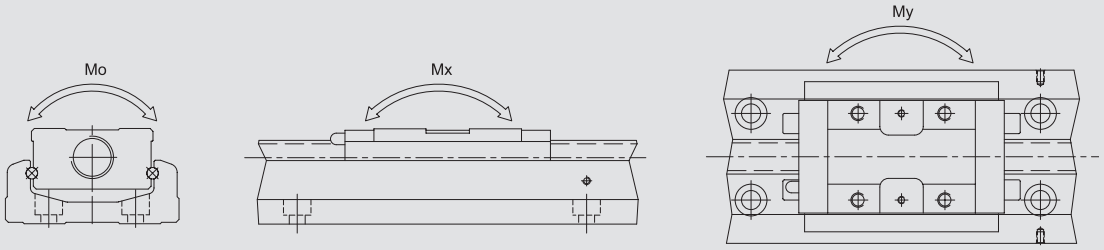
## 6.2 KK Linear Stages – Technical Data

### 6.2.1 Maximum Speeds of the KK Modules

| Model | Ballscrew Lead [mm] | Rail length [mm] | Speed [mm/s] Precision | Standard |
|---|---|---|---|---|
| **KK40** | 01 | 100 | 190 | – |
| | 01 | 150 | 190 | – |
| | 01 | 200 | 190 | – |
| **KK50** | 02 | 150 | 270 | – |
| | 02 | 200 | 270 | – |
| | 02 | 250 | 270 | – |
| | 02 | 300 | 270 | – |
| **KK60** | 05 | 150 | 550 | 390 |
| | 05 | 200 | 550 | 390 |
| | 05 | 300 | 550 | 390 |
| | 05 | 400 | 550 | 390 |
| | 05 | 500 | 550 | 390 |
| | 05 | 600 | 340 | 340 |
| **KK60** | 10 | 150 | 1100 | 790 |
| | 10 | 200 | 1100 | 790 |
| | 10 | 300 | 1100 | 790 |
| | 10 | 400 | 1100 | 790 |
| | 10 | 500 | 1100 | 790 |
| | 10 | 600 | 670 | 670 |
| **KK86** | 10 | 340 | 740 | 520 |
| | 10 | 440 | 740 | 520 |
| | 10 | 540 | 740 | 520 |
| | 10 | 640 | 740 | 520 |
| | 10 | 740 | 740 | 520 |
| | 10 | 940 | – | 430 |
| **KK86** | 20 | 340 | 1480 | 1050 |
| | 20 | 440 | 1480 | 1050 |
| | 20 | 540 | 1480 | 1050 |
| | 20 | 640 | 1480 | 1050 |
| | 20 | 740 | 1480 | 1050 |
| | 20 | 940 | – | 870 |
| **KK100** | 20 | 980 | 1120 | – |
| | 20 | 1080 | 980 | – |
| | 20 | 1180 | 750 | – |
| | 20 | 1280 | 490 | – |
| | 20 | 1380 | 425 | – |

## 6.2.2 Load Capacities

### Display of Static Moments Affecting the KK Stages




### Load Capacity of KK Stages

| | | KK4001 | KK5002 | KK6005 | | KK6010 | | KK8610 | | KK8620 | | KK10020 | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | P* | P* | P* | C** | P* | C** | P* | C** | P* | C** | P* | C** |
| Ballscrew | | | | | | | | | | | | | |
| Nominal diameter [mm] | | 8 | 8 | 12 | 12 | 12 | 12 | 15 | 15 | 15 | 15 | 20 | 20 |
| Lead [mm] | | 1 | 2 | 5 | 5 | 10 | 10 | 10 | 10 | 20 | 20 | 20 | 20 |
| Dynamic load [N] | | 735 | 2136 | 3744 | 3377 | 2410 | 2107 | 7144 | 6429 | 4645 | 4175 | 7046 | 4782 |
| Static load [N] | | 1538 | 3489 | 6243 | 5625 | 3743 | 3234 | 12642 | 11387 | 7655 | 6889 | 12544 | 9163 |
| Linear guideway | | | | | | | | | | | | | |
| Dynamic load [N] | Standard block A | 3920 | 8007 | 13230 | 13230 | 13230 | 13230 | 31458 | 31458 | 31458 | 31458 | 39200 | 39200 |
| | Short block S | – | – | 7173 | 7173 | 7173 | 7173 | – | – | – | – | – | – |
| Static load [N] | Standard block A | 6468 | 12916 | 21462 | 21462 | 21462 | 21462 | 50764 | 50764 | 50764 | 50764 | 63406 | 63406 |
| | Short block S | – | – | 11574 | 11574 | 11574 | 11574 | – | – | – | – | – | – |
| Permissible static moment Mx Pitching [N-m] | Standard block A1 | 33 | 116 | 152 | 152 | 152 | 152 | 622 | 622 | 622 | 622 | 960 | 960 |
| | Standard block A2 | 182 | 278 | 348 | 348 | 348 | 348 | 3050 | 3050 | 3050 | $30_{50}$ | 4763 | 4763 |
| | Short block S1 | – | – | 72 | 72 | 72 | 72 | – | – | – | – | – | – |
| | Short block S2 | – | – | 205 | 205 | 205 | 205 | – | – | – | – | – | – |
| Permissible static moment My Yawing [N-m] | Standard block A1 | 33 | 116 | 152 | 152 | 152 | 152 | 622 | 622 | 622 | 622 | 960 | 960 |
| | Standard block A2 | 182 | 278 | 348 | 348 | 348 | 348 | 3050 | 3050 | 3050 | 3050 | 4763 | 4763 |
| | Short block S1 | – | – | 72 | 72 | 72 | 72 | – | – | – | – | – | – |
| | Short block S2 | – | – | 205 | 205 | 205 | 205 | – | – | – | – | – | – |
| Permissible static moment Mo Rolling [N-m] | Standard block A1 | 81 | 222 | 419 | 419 | 419 | 419 | 1507 | 1507 | 1507 | 1507 | 2205 | 2205 |
| | Standard block A2 | 162 | 444 | 838 | 838 | 838 | 838 | 3014 | 3014 | 3014 | 3014 | 4410 | 4410 |
| | Short block S1 | - | - | 241 | 241 | 241 | 241 | - | - | - | - | - | - |
| | Short block S2 | - | - | 482 | 482 | 482 | 482 | - | - | | | | |

* P = Precision KK stage
** C = Standard KK stage

### 6.2.3 Accuracies

**Accuracies for KK Stages**

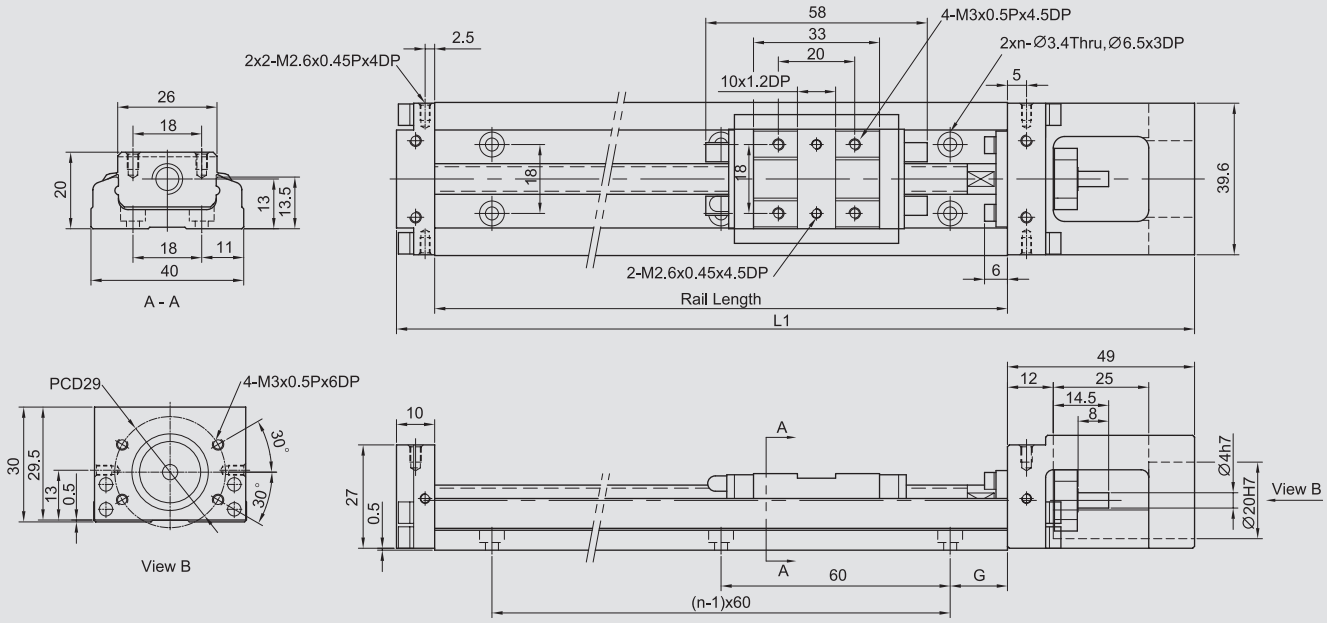
| Type | Rail length [mm] | Repeatability [mm] | | Accuracy [mm] | | Guideway parallelism [mm] | | Starting torque [Nmm] | |
|---|---|---|---|---|---|---|---|---|---|
| | | P* | C** | P* | C** | P* | C** | P* | C** |
| KK40 | 100 | ±0,003 | - | 0,020 | - | 0,010 | - | 12 | - |
| | 150 | ±0,003 | - | 0,020 | - | 0,010 | - | 12 | - |
| | 200 | ±0,003 | - | 0,020 | - | 0,010 | - | 12 | - |
| KK50 | 150 | ±0,003 | - | 0,020 | - | 0,010 | - | 40 | - |
| | 200 | ±0,003 | - | 0,020 | - | 0,010 | - | 40 | - |
| | 250 | ±0,003 | - | 0,020 | - | 0,010 | - | 40 | - |
| | 300 | ±0,003 | - | 0,020 | - | 0,010 | - | 40 | - |
| KK60 | 150 | ±0,003 | ±0,01 | 0,020 | - | 0,010 | - | 150 | 70 |
| | 200 | ±0,003 | ±0,01 | 0,020 | - | 0,010 | - | 150 | 70 |
| | 300 | ±0,003 | ±0,01 | 0,020 | - | 0,010 | - | 150 | 70 |
| | 400 | ±0,003 | ±0,01 | 0,020 | - | 0,010 | - | 150 | 70 |
| | 500 | ±0,003 | ±0,01 | 0,020 | - | 0,010 | - | 150 | 70 |
| | 600 | ±0,003 | ±0,01 | 0,020 | - | 0,010 | - | 150 | 70 |
| KK86 | 340 | ±0,003 | ±0,01 | 0,025 | - | 0,015 | - | 150 | 100 |
| | 440 | ±0,003 | ±0,01 | 0,025 | - | 0,015 | - | 150 | 100 |
| | 540 | ±0,003 | ±0,01 | 0,025 | - | 0,015 | - | 150 | 100 |
| | 640 | ±0,003 | ±0,01 | 0,025 | - | 0,015 | - | 150 | 100 |
| | 740 | ±0,003 | ±0,01 | 0,030 | - | 0,020 | - | 170 | 100 |
| | 940 | ±0,003 | ±0,01 | 0,040 | - | 0,030 | - | 250 | 100 |
| KK100 | 980 | ±0,005 | ±0,01 | 0,035 | - | 0,025 | - | 170 | 120 |
| | 1080 | ±0,005 | ±0,01 | 0,035 | - | 0,025 | - | 170 | 120 |
| | 1180 | ±0,005 | ±0,01 | 0,040 | - | 0,030 | - | 200 | 120 |
| | 1280 | ±0,005 | ±0,01 | 0,045 | - | 0,030 | - | 230 | 150 |
| | 1380 | ±0,005 | ±0,01 | 0,050 | - | 0,040 | - | 250 | 150 |

* P = Precision KK stage
** C = Standard KK stage

**Reference Side**
When observed from the motor flange, the reference side is located on the left side of the linear module

## 6.2.4 Dimensions of KK40 Stages

### KK40 Stages without Cover

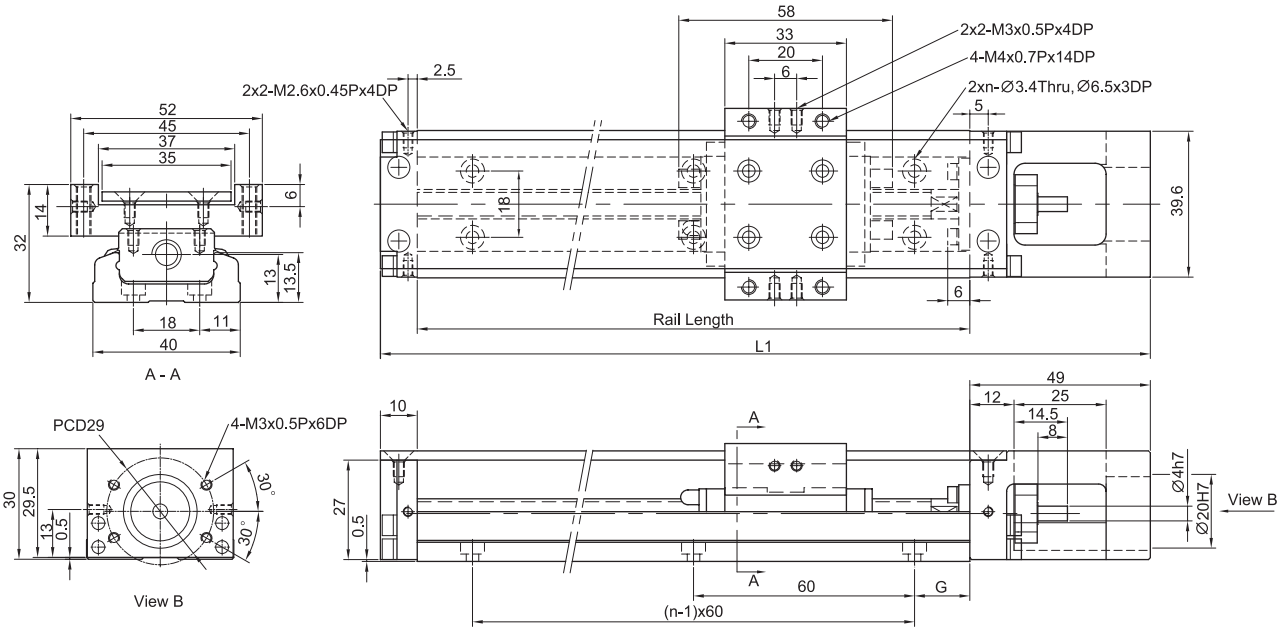



### Dimensions and Mass of the KK40 Stages without Cover

| | | Rail length | | |
|---|---|---|---|---|
| | | 100 | 150 | 200 |
| Total length L1 [mm] | | 159 | 209 | 259 |
| Max. stroke [mm] | Block A1 | 36 | 86 | 136 |
| | Block A2 | – | 34 | 84 |
| G [mm] | | 20 | 15 | 40 |
| n | | 2 | 3 | 3 |
| Mass [kg] | Block A1 | 0,48 | 0,6 | 0,72 |
| | Block A2 | – | 0,67 | 0,79 |

# Positioning Systems
## Linear Module

### KK40 Stages with Aluminium Cover



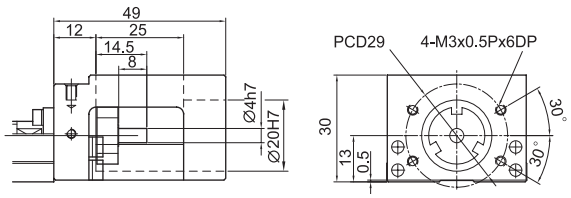

### Dimensions and Mass of the KK40 Stages with AluminiumCover

| | | Rail length | | |
|---|---|---|---|---|
| | | 100 | 150 | 200 |
| Total length L1 [mm] | | 209 | 259 | 159 |
| Max. stroke [mm] | Block A1 | 86 | 136 | 36 |
| | Block A2 | – | 34 | 84 |
| G [mm] | | 20 | 15 | 40 |
| n | | 2 | 3 | 3 |
| Mass [kg] | Block A1 | 0,55 | 0,68 | 0,82 |
| | Block A2 | – | 0,76 | 0,89 |

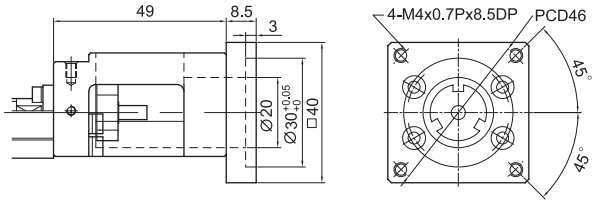
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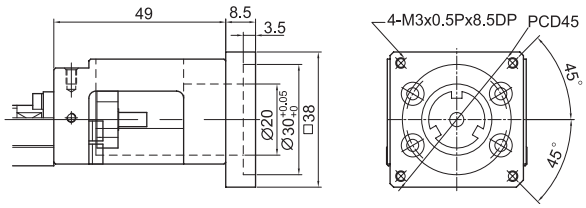
## KK40 Stages Adapter Flange F0



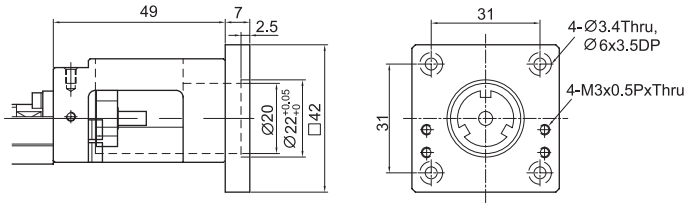

## KK40 Stages Adapter Flange F1




## KK40 Stages Adapter Flange F2



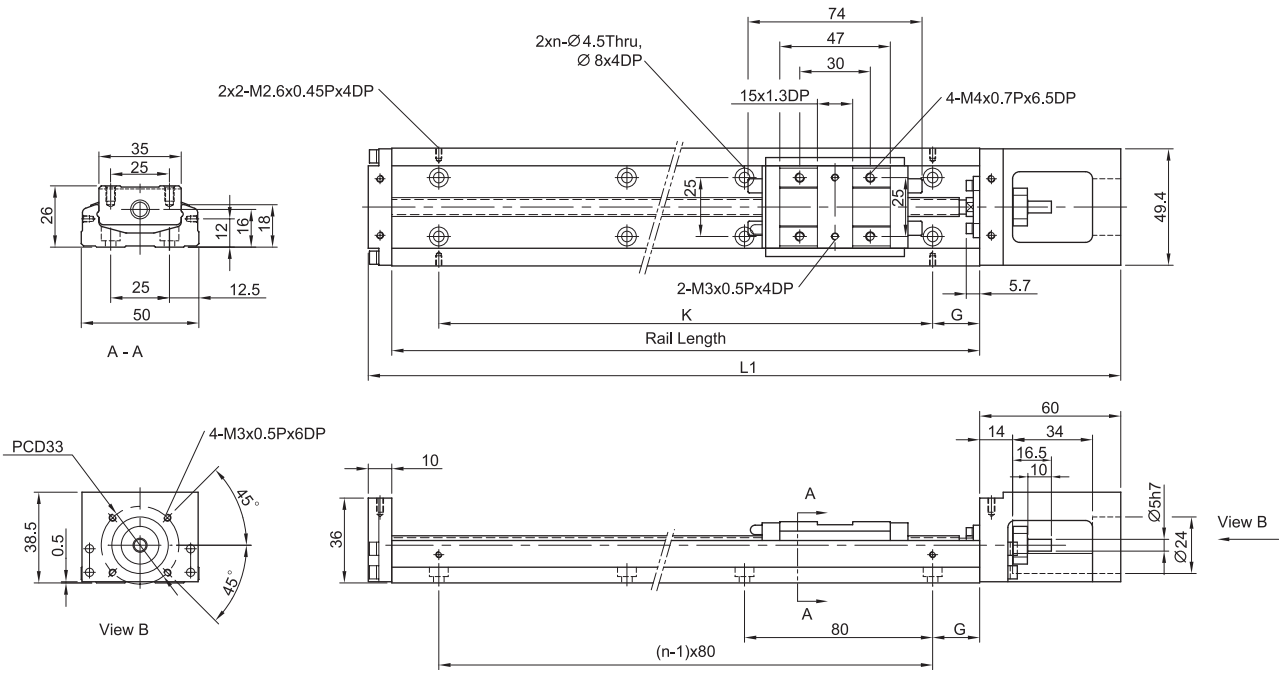

## KK40 Stages Adapter Flange F3



# Positioning Systems
Linear Module

## 6.2.5 Dimensions of KK Stages KK50

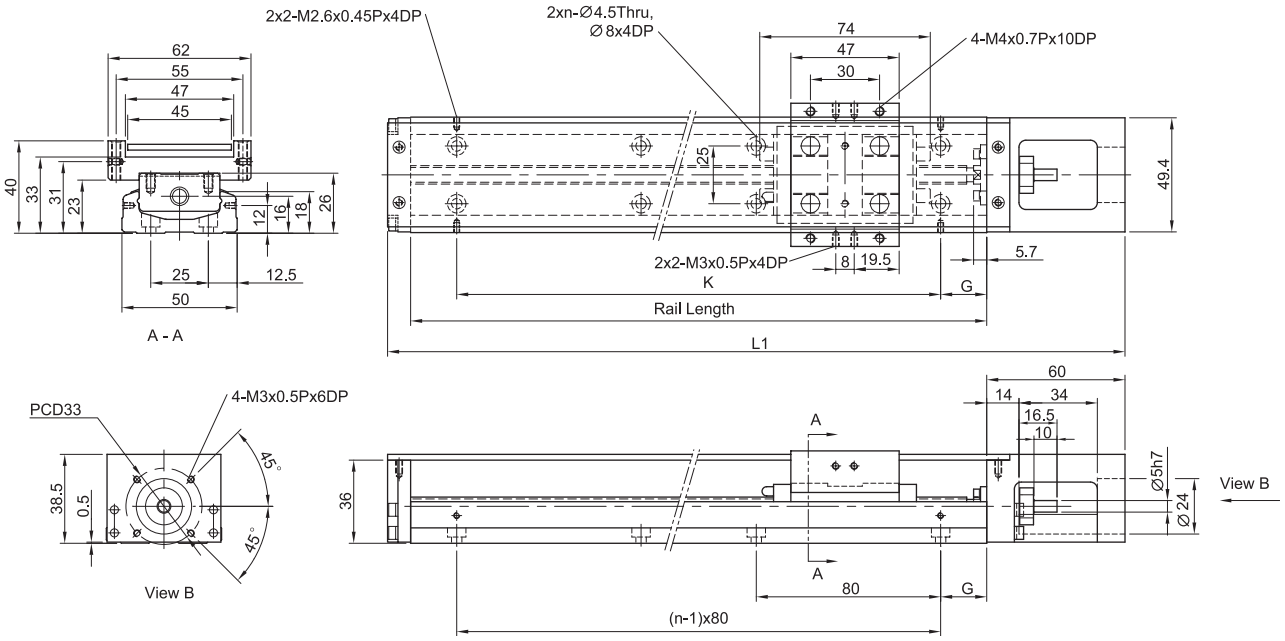
### KK50 Stage without Cover




### Dimensions and Mass of the KK50-Stages without Cover

| Rail length [mm] | Total length L1 [mm] | Maximum stroke [mm] Block A1 | Block A2 | G [mm] | K [mm] | n | Mass [kg] Block A1 | Block A2 |
|---|---|---|---|---|---|---|---|---|
| 150 | 220 | 70 | - | 35 | 80 | 2 | 1 | - |
| 200 | 270 | 120 | 55 | 20 | 160 | 3 | 1,2 | 1,4 |
| 250 | 320 | 170 | 105 | 45 | 160 | 3 | 1,4 | 1,6 |
| 300 | 370 | 220 | 155 | 30 | 240 | 4 | 1,6 | 1,8 |

## KK50 Stages with Aluminium Cover

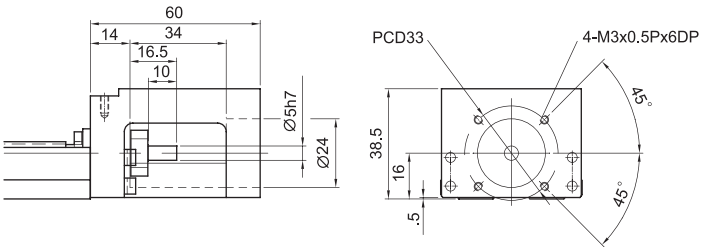



## Dimensions and Mass of the KK50-Stages with Aluminium Cover

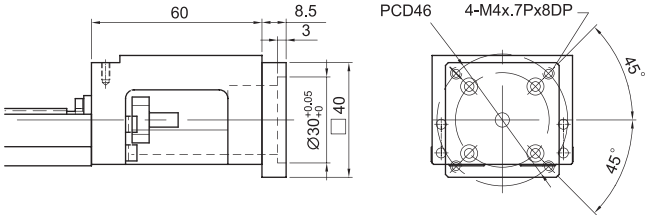
| Rail length [mm] | Total length L1 [mm] | Maximum stroke [mm] Block A1 | Block A2 | G [mm] | K [mm] | n | Mass [kg] Block A1 | Block A2 |
|---|---|---|---|---|---|---|---|---|
| 150 | 220 | 70 | - | 35 | 80 | 2 | 1,1 | - |
| 200 | 270 | 120 | 55 | 20 | 160 | 3 | 1,3 | 1,5 |
| 250 | 320 | 170 | 105 | 45 | 160 | 3 | 1,6 | 1,8 |
| 300 | 370 | 220 | 155 | 30 | 240 | 4 | 1,8 | 2,0 |

# Positioning Systems
## Linear Module

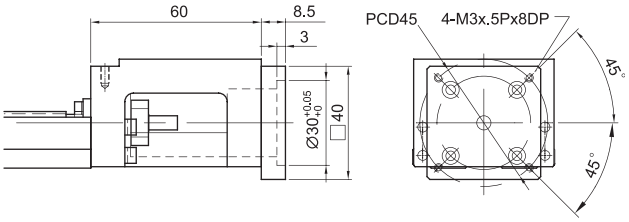
### KK50 Stages Adapter Flange F0



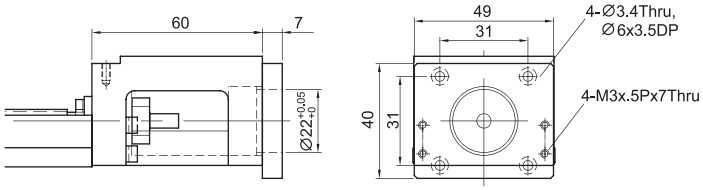

### KK50 Stages Adapter Flange F1



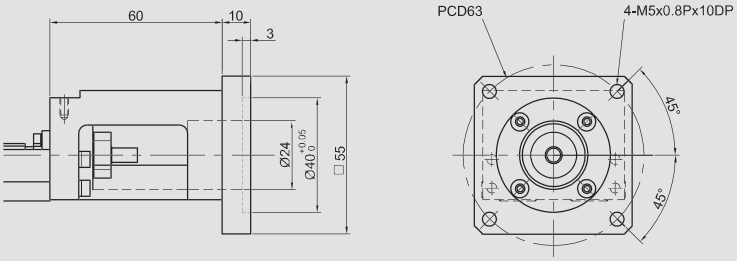

### KK50 Stages Adapter Flange F2



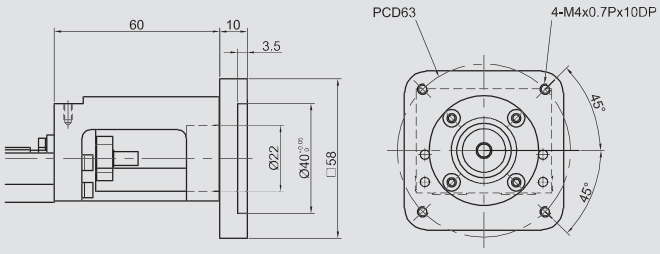

### KK50 Stages Adapter Flange F3




### KK50 Stages Adapter Flange F4



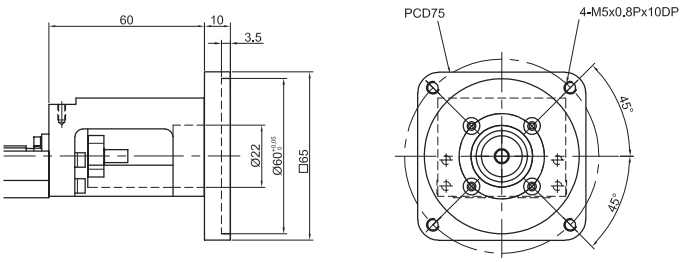

### KK50 Stages Adapter Flange F5



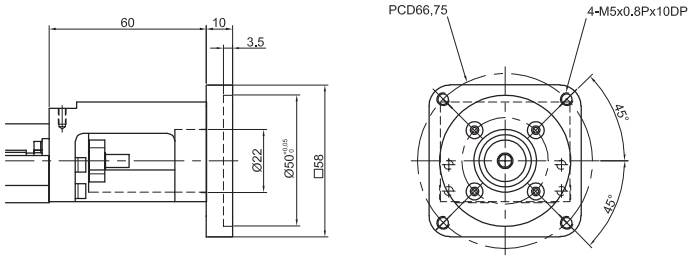
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## KK50 Stages Adapter Flange F6

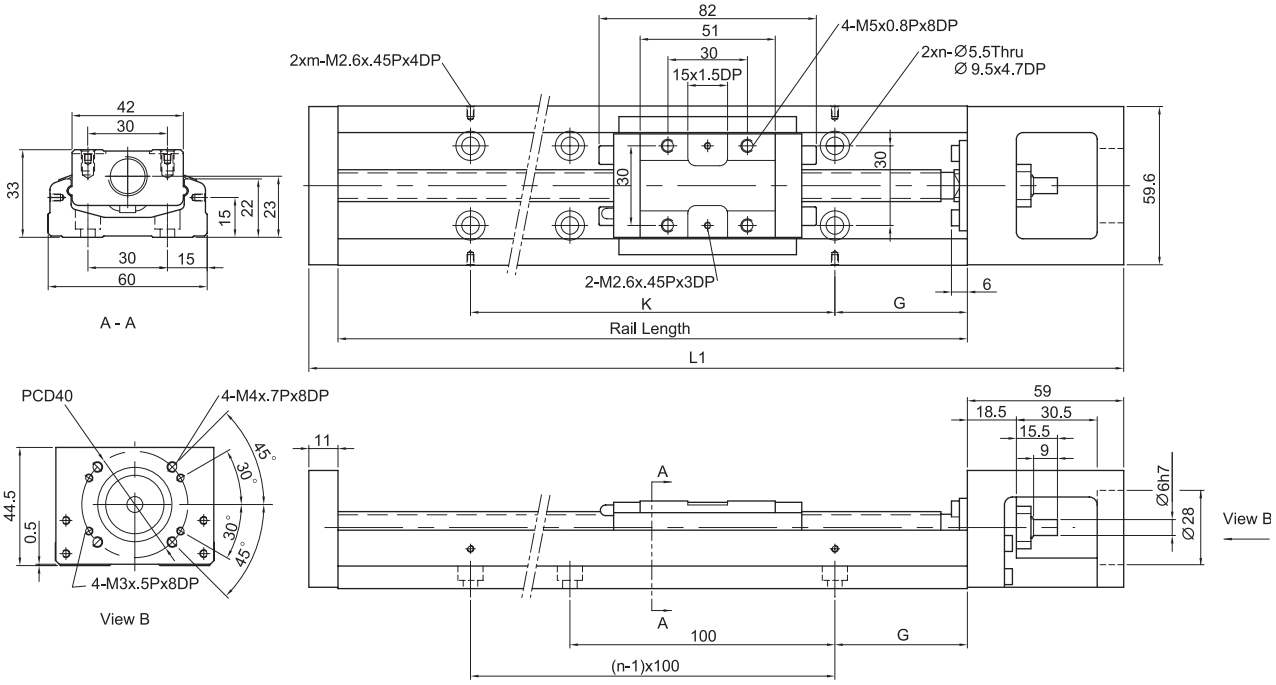



## KK50 Stages Adapter Flange F7



### 6.2.6 Dimensions of KK Stages KK50

**KK60 Stages without Cover, Standard Block**



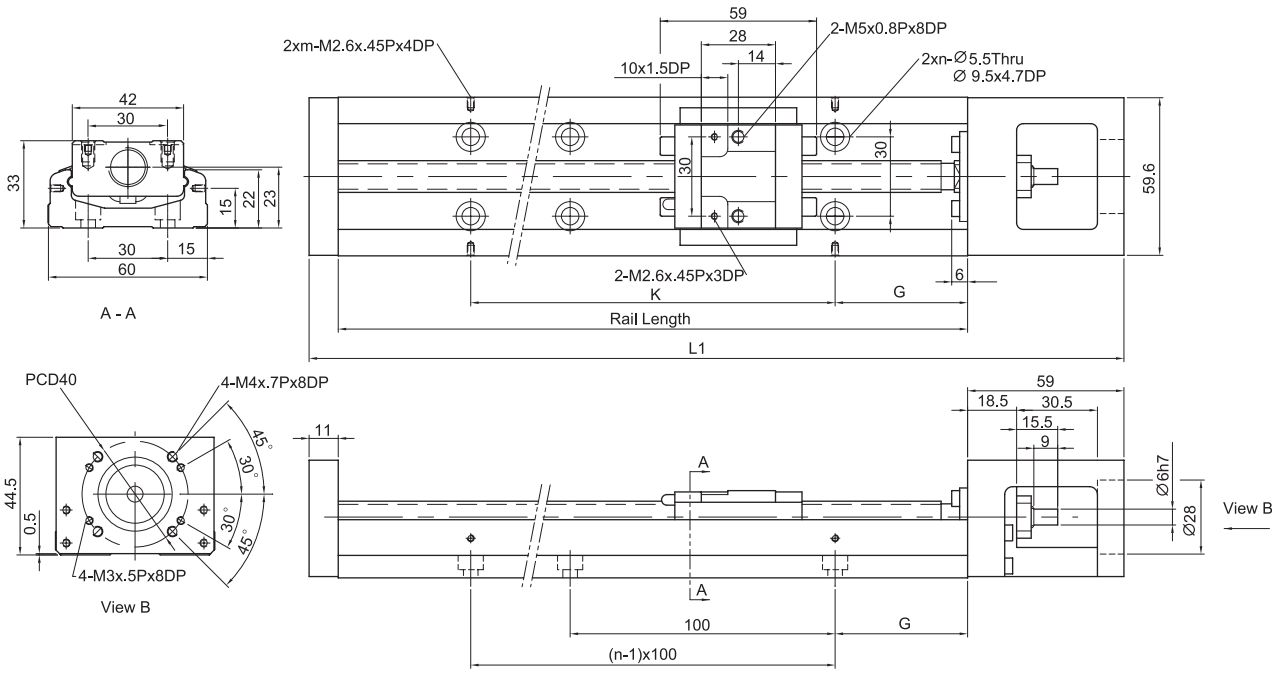

**Dimensions and Mass of the KK60 Stages without Cover, Standard Block**

| Rail length [mm] | Total length L1 [mm] | Maximum stroke [mm] Block A1 | Block A2 | G [mm] | K [mm] | n | m | Mass [kg] Block A1 | Block A2 |
|---|---|---|---|---|---|---|---|---|---|
| 150 | 220 | 60 | - | 25 | 100 | 2 | 2 | 1,5 | - |
| 200 | 270 | 110 | - | 50 | 100 | 2 | 2 | 1,8 | - |
| 300 | 370 | 210 | 135 | 50 | 200 | 3 | 2 | 2,4 | 2,7 |
| 400 | 470 | 310 | 235 | 50 | 100 | 4 | 4 | 3 | 3,3 |
| 500 | 570 | 410 | 335 | 50 | 200 | 5 | 3 | 3,6 | 3,9 |
| 600 | 670 | 510 | 435 | 50 | 100 | 6 | 6 | 4,2 | 4,6 |

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## KK60 Stages without Cover, Short Block

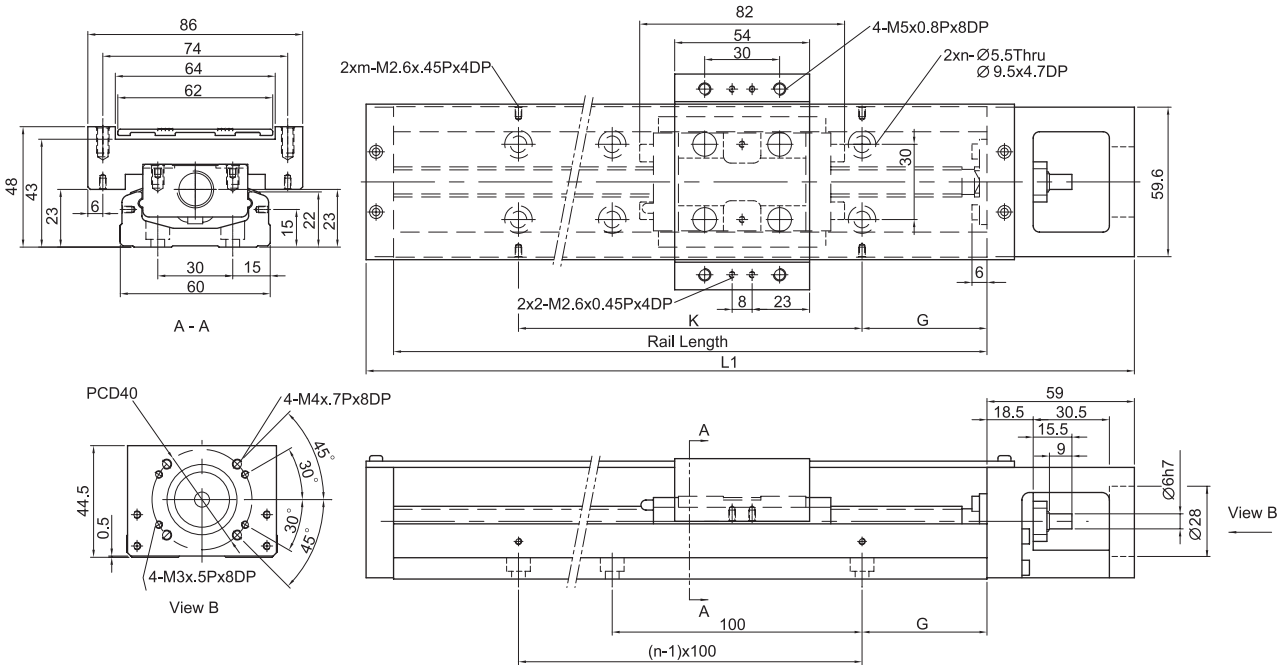



## Dimensions and Mass of the KK60 Stages without Cover, Short Block

| Rail length [mm] | Total length L1 [mm] | Maximum stroke [mm] Block A1 | Block A2 | G [mm] | K [mm] | n | m | Mass [kg] Block A1 | Block A2 |
|---|---|---|---|---|---|---|---|---|---|
| 150 | 220 | 85 | 34 | 25 | 100 | 2 | 2 | 1,4 | 1,6 |
| 200 | 270 | 135 | 84 | 50 | 100 | 2 | 2 | 1,7 | 1,9 |
| 300 | 370 | 235 | 184 | 50 | 200 | 3 | 2 | 2,3 | 2,5 |
| 400 | 470 | 335 | 284 | 50 | 100 | 4 | 4 | 2,9 | 3,1 |
| 500 | 570 | 435 | 384 | 50 | 200 | 5 | 3 | 3,5 | 3,7 |
| 600 | 670 | 535 | 484 | 50 | 100 | 6 | 6 | 4,1 | 4,3 |

# Positioning Systems
## Linear Module

### KK60 Stages with Aluminium Cover, Standard Block



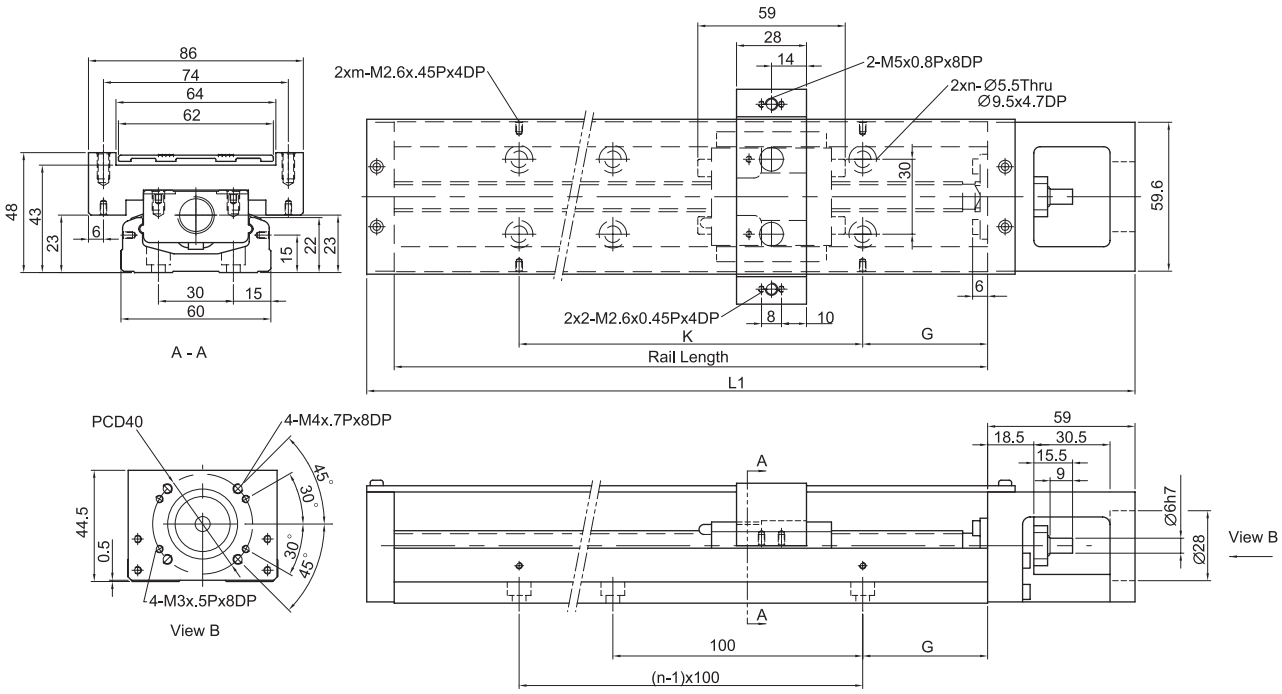

### Dimensions and Mass of the KK60 Stages without Cover, Short Block

| Rail length [mm] | Total length L1 [mm] | Maximum stroke [mm] Block A1 | Block A2 | G [mm] | K [mm] | n | m | Mass [kg] Block A1 | Block A2 |
|---|---|---|---|---|---|---|---|---|---|
| 150 | 220 | 60 | - | 25 | 100 | 2 | 2 | 1,7 | - |
| 200 | 270 | 110 | - | 50 | 100 | 2 | 2 | 2,1 | - |
| 300 | 370 | 210 | 135 | 50 | 200 | 3 | 2 | 2,7 | 3,0 |
| 400 | 470 | 310 | 235 | 50 | 100 | 4 | 4 | 3,3 | 3,6 |
| 500 | 570 | 410 | 335 | 50 | 200 | 5 | 3 | 3,9 | 4,2 |
| 600 | 670 | 510 | 435 | 50 | 100 | 6 | 6 | 4,6 | 5,0 |

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## KK60 Stages with Aluminium Cover, Short Block

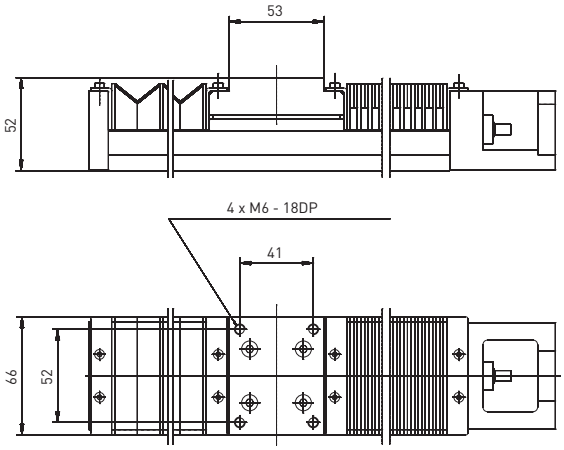



## Dimensions and Mass of the KK60-Stages with Aluminium Cover, Short Block

| Rail length [mm] | Total length L1 [mm] | Maximum stroke [mm] Block A1 | Block A2 | G [mm] | K [mm] | n | m | Mass [kg] Block A1 | Block A2 |
|---|---|---|---|---|---|---|---|---|---|
| 150 | 220 | 85 | 34 | 25 | 100 | 2 | 2 | 1,6 | 1,8 |
| 200 | 270 | 135 | 84 | 50 | 100 | 2 | 2 | 1,9 | 2,1 |
| 300 | 370 | 235 | 184 | 50 | 200 | 3 | 2 | 2,5 | 2,7 |
| 400 | 470 | 335 | 284 | 50 | 100 | 4 | 4 | 3,1 | 3,3 |
| 500 | 570 | 435 | 384 | 50 | 200 | 5 | 3 | 3,7 | 3,9 |
| 600 | 670 | 535 | 484 | 50 | 100 | 6 | 6 | 4,4 | 4,6 |

# Positioning Systems
## Linear Module

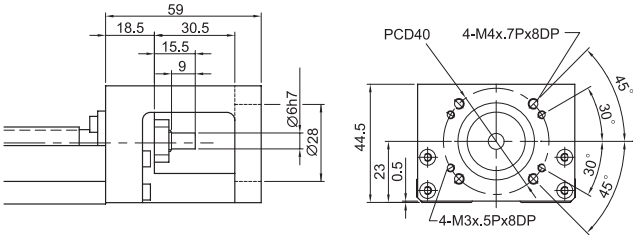
### KK60 Stages with Bellow Cover



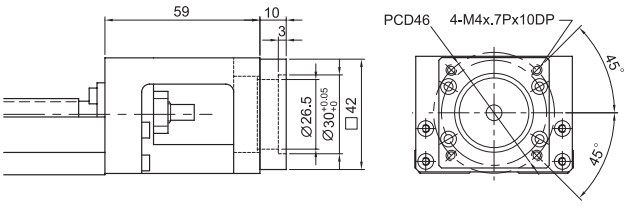

### Dimensions and Mass of the KK60 Stages with Bellow Cover

| Rail length [mm] | Mass [kg] | Maximum stroke [mm] Block A1 | Block A2 |
|---|---|---|---|
| 150 | 1,7 | 45 | – |
| 200 | 2,1 | 77 | – |
| 300 | 2,7 | 151 | 93 |
| 400 | 3,3 | 230 | 165 |
| 500 | 3,9 | 300 | 241 |
| 600 | 4,6 | 376 | 317 |

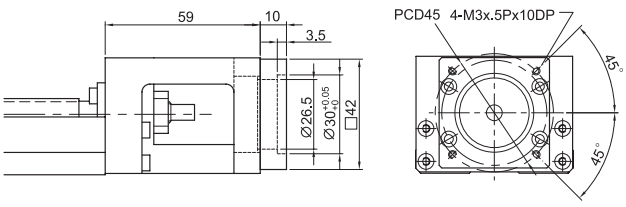
### KK60 Stages Adapter Flange F0



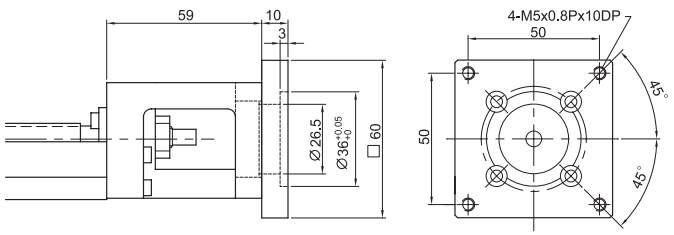

### KK60 Stages Adapter Flange F1




### KK60 Stages Adapter Flange F2



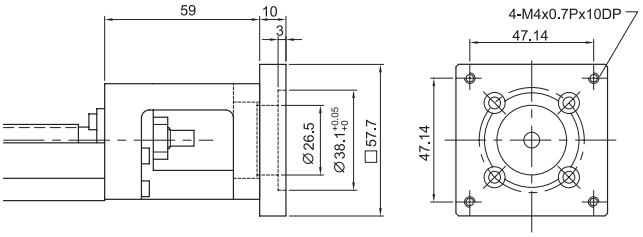

### KK60 Stages Adapter Flange F3



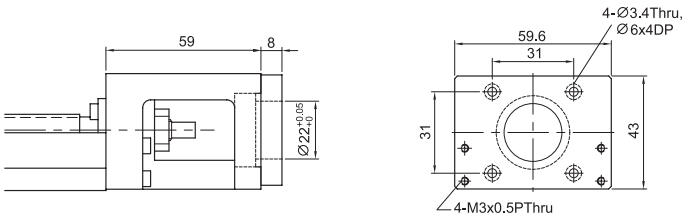
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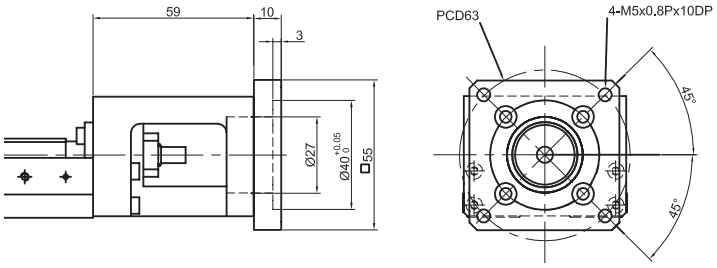
## KK60 Stages Adapter Flange F4



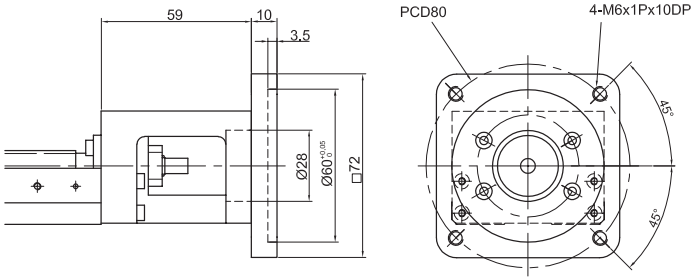

## KK60 Stages Adapter Flange F5



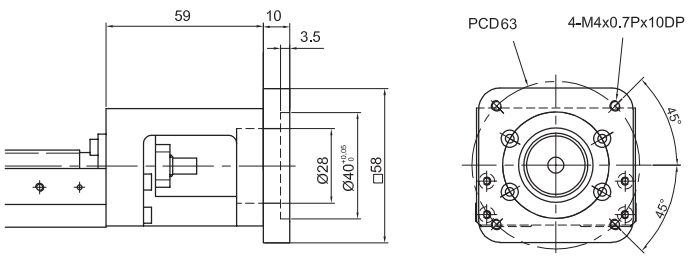

## KK60 Stages Adapter Flange F6



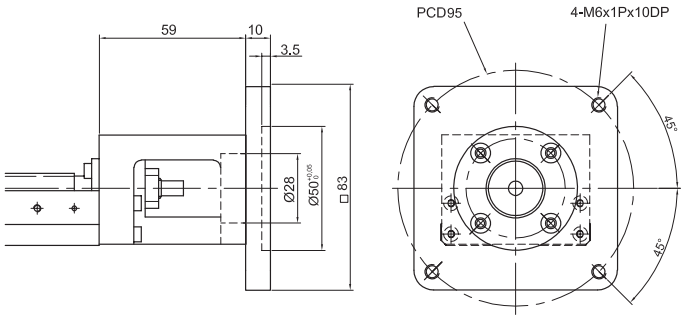

## KK60 Stages Adapter Flange F7



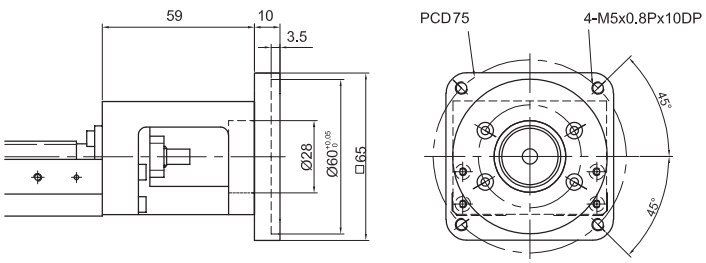

## KK60 Stages Adapter Flange F8



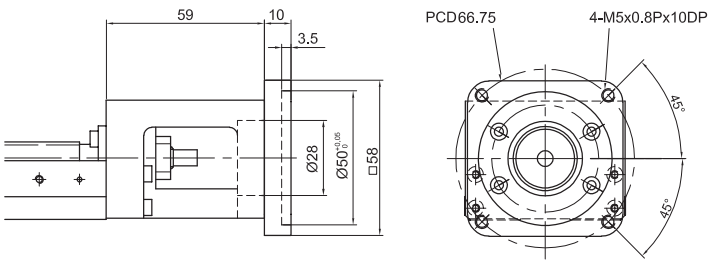

## KK60 Stages Adapter Flange F9




## KK60 Stages Adapter Flange F10

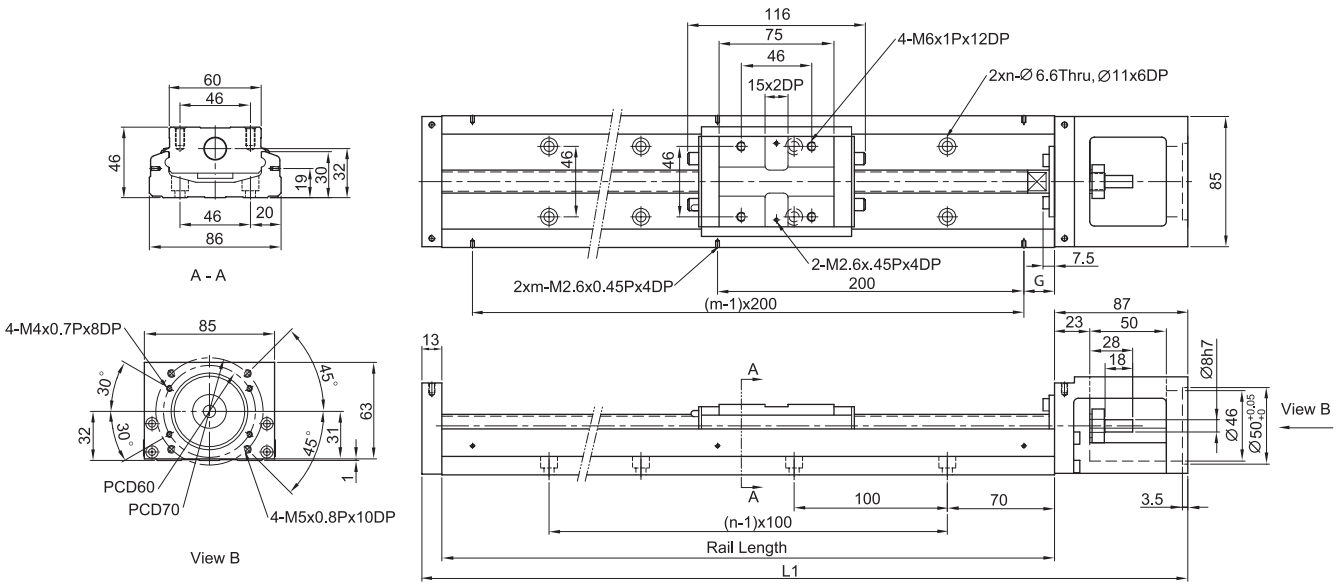



## KK60 Stages Adapter Flange F11



### 6.2.7 Dimensions of KK Stages KK86

**KK86 Stages without Cover**



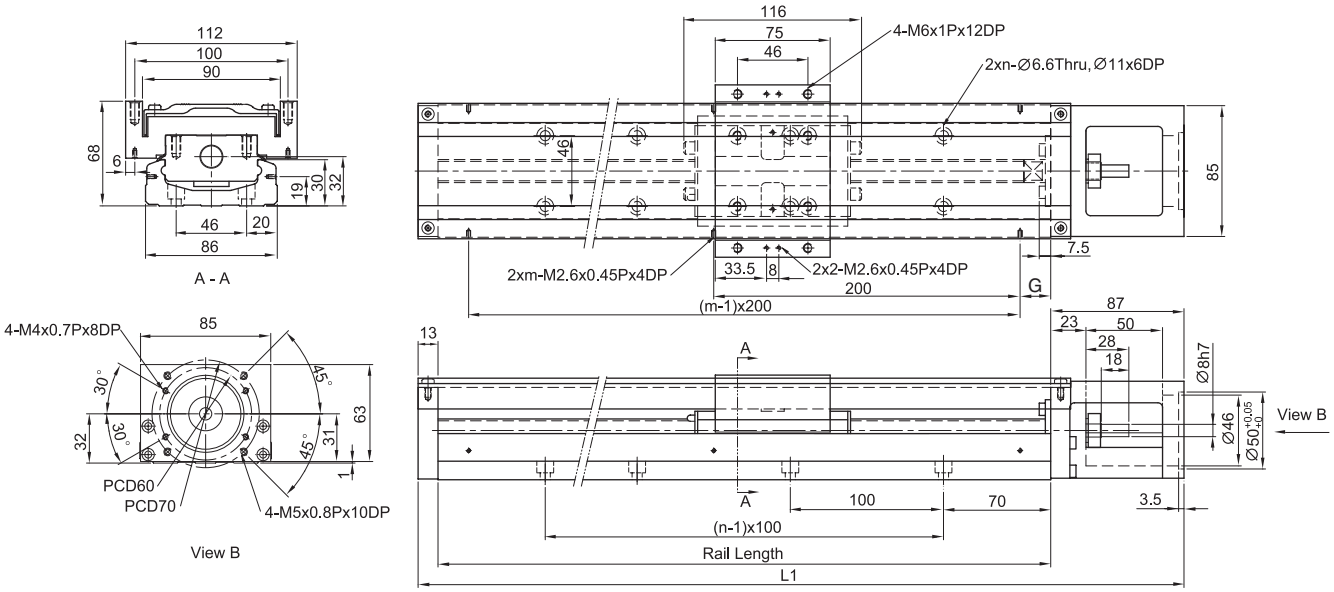

**Dimensions and Mass of the KK86 Stages without Cover**

| Rail length [mm] | Total length L1 [mm] | Maximum stroke [mm] Block A1 | Block A2 | G [mm] | n | m | Mass [kg] Block A1 | Block A2 |
|---|---|---|---|---|---|---|---|---|
| 340 | 440 | 210 | 100 | 70 | 3 | 2 | 5,7 | 6,5 |
| 440 | 540 | 310 | 200 | 20 | 4 | 3 | 6,9 | 7,7 |
| 540 | 640 | 410 | 300 | 70 | 5 | 3 | 8,0 | 8,8 |
| 640 | 740 | 510 | 400 | 30 | 6 | 4 | 9,2 | 10,0 |
| 740 | 840 | 610 | 500 | 70 | 7 | 4 | 10,4 | 11,2 |
| 940 | 1040 | 810 | 700 | 70 | 9 | 5 | 11,6 | 12,4 |

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## KK86 Stages with Aluminium Cover

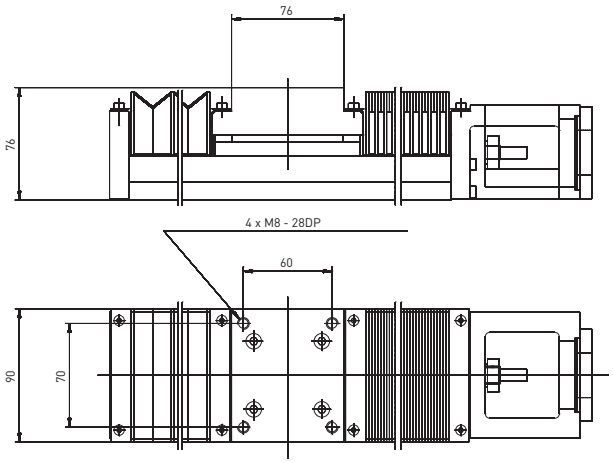



## Dimensions and Mass of the KK86 Stages with Aluminium Cover

| Rail length [mm] | Total length L1 [mm] | Maximum stroke [mm] Block A1 | Block A2 | G [mm] | n | m | Mass [kg] Block A1 | Block A2 |
|---|---|---|---|---|---|---|---|---|
| 340 | 440 | 210 | 100 | 70 | 3 | 2 | 6,5 | 7,3 |
| 440 | 540 | 310 | 200 | 20 | 4 | 3 | 7,8 | 8,6 |
| 540 | 640 | 410 | 300 | 70 | 5 | 3 | 9,0 | 9,8 |
| 640 | 740 | 510 | 400 | 30 | 6 | 4 | 10,3 | 11,3 |
| 740 | 840 | 610 | 500 | 70 | 7 | 4 | 11,6 | 12,4 |
| 940 | 1040 | 810 | 700 | 70 | 9 | 5 | 13,0 | 13,8 |

# Positioning Systems
## Linear Module

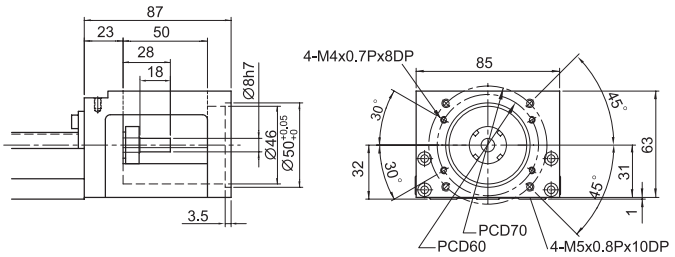
### KK86 Stages with Bellow Cover



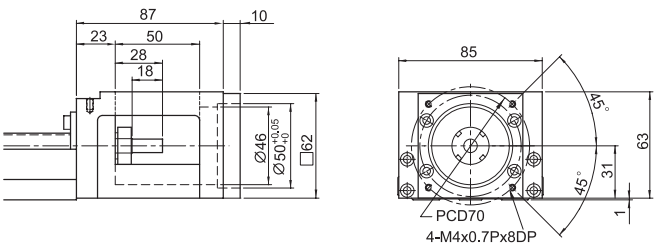

### Dimensions and Mass of the KK86 Stages with Bellow Cover

| Rail length [mm] | Mass [kg] | Maximum stroke [mm] Block A1 | Block A2 |
|---|---|---|---|
| 340 | 6,3 | 174 | 84 |
| 440 | 7,6 | 248 | 158 |
| 540 | 8,8 | 327 | 237 |
| 640 | 10 | 410 | 318 |
| 740 | 11,3 | 491 | 399 |
| 940 | 12,7 | 654 | 561 |

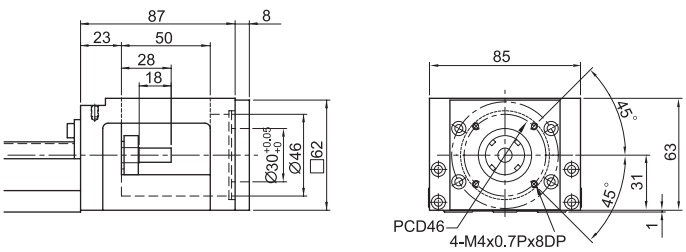
### KK86 Stages Adapter Flange F0



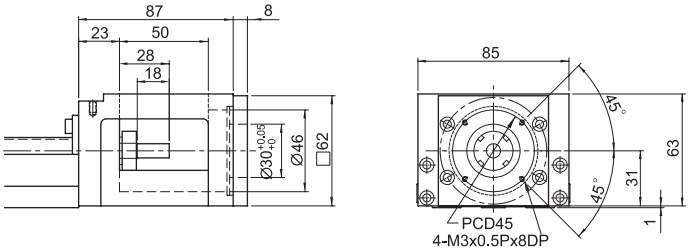

### KK86 Stages Adapter Flange F1




### KK86 Stages Adapter Flange F2



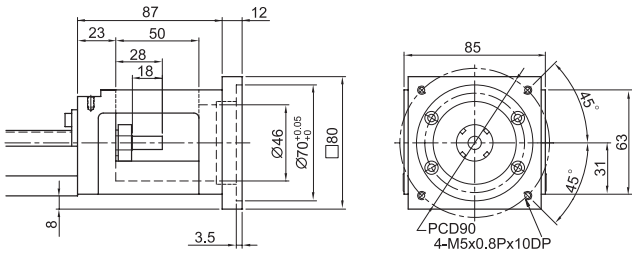

### KK86 Stages Adapter Flange F3



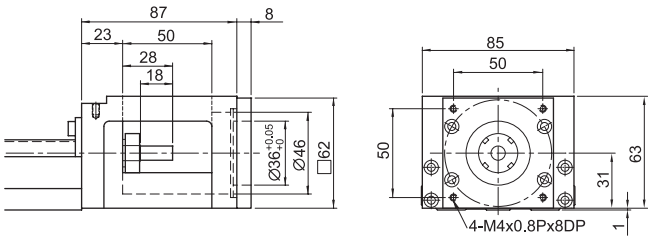
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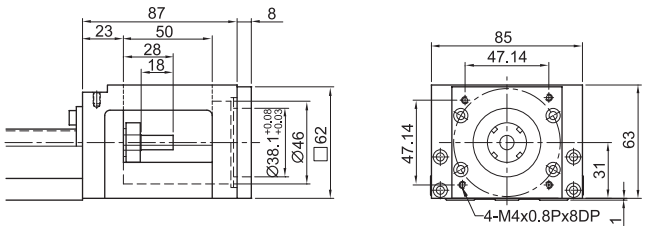
## KK86 Stages Aapter Flange F4



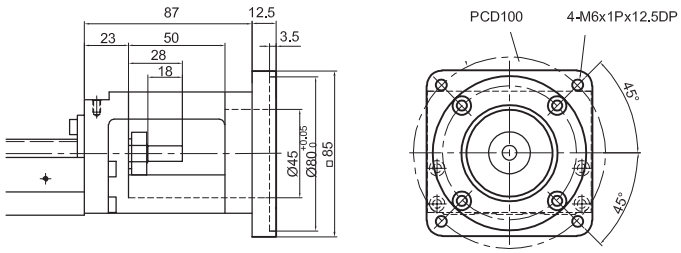

## KK86 Stages Adapter Flange F5



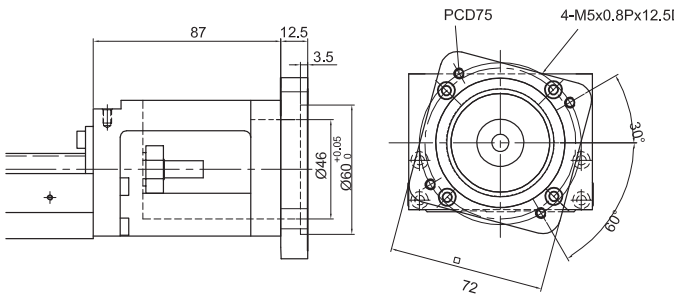

## KK86 Stages Adapter Flange F6



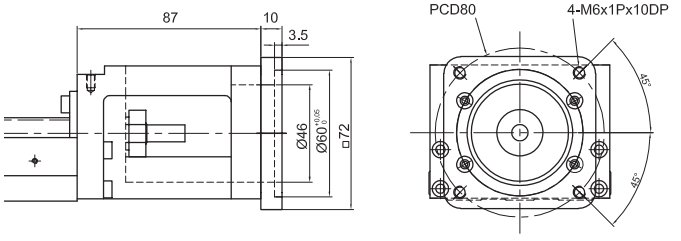

## KK86 Stages Adapter Flange F7



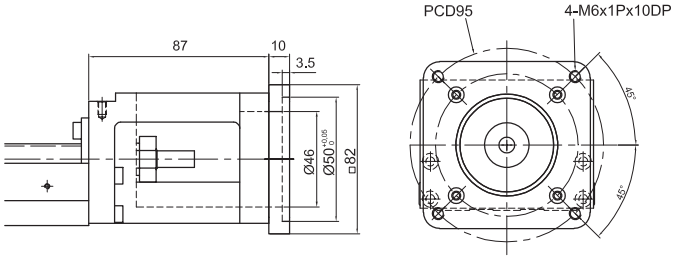

## KK86 Stages Adapter Flange F8




## KK86 Stages Adapter Flange F9

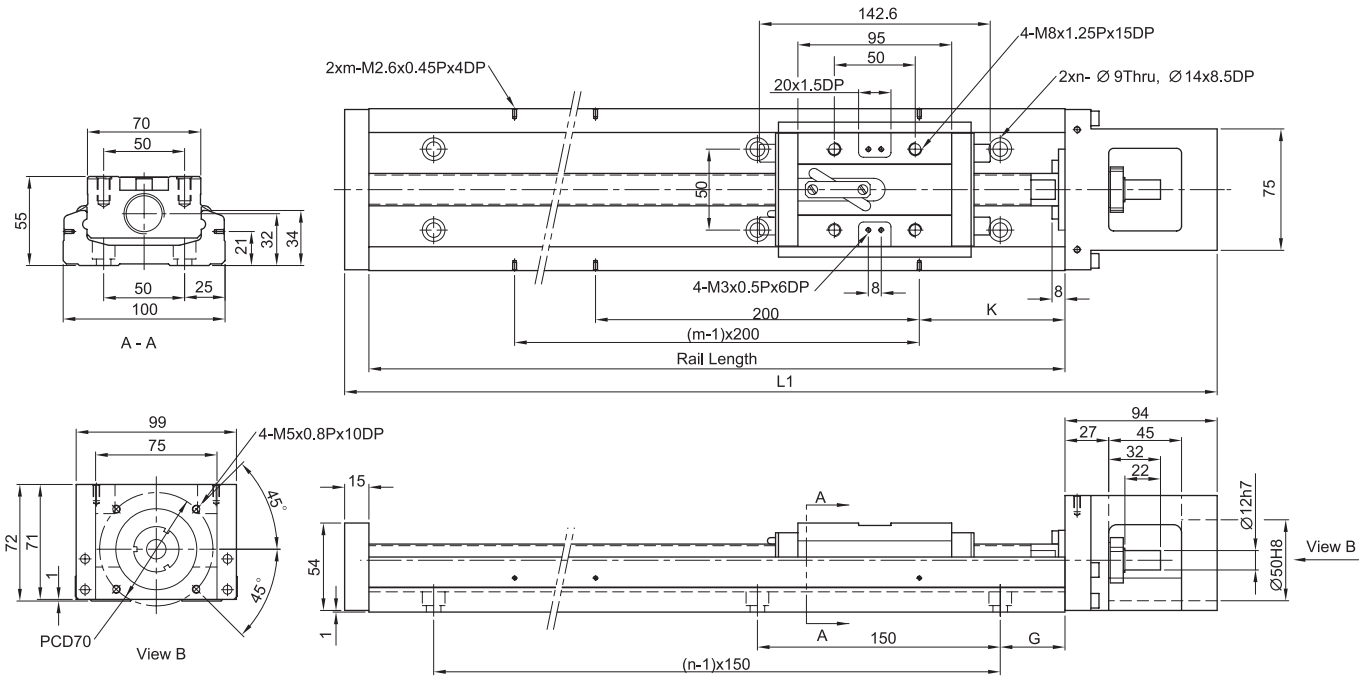



## KK86 Stages Adapter Flange F10



### 6.2.8 Dimensions of KK Stage KK100

**KK100 Stages without Cover**



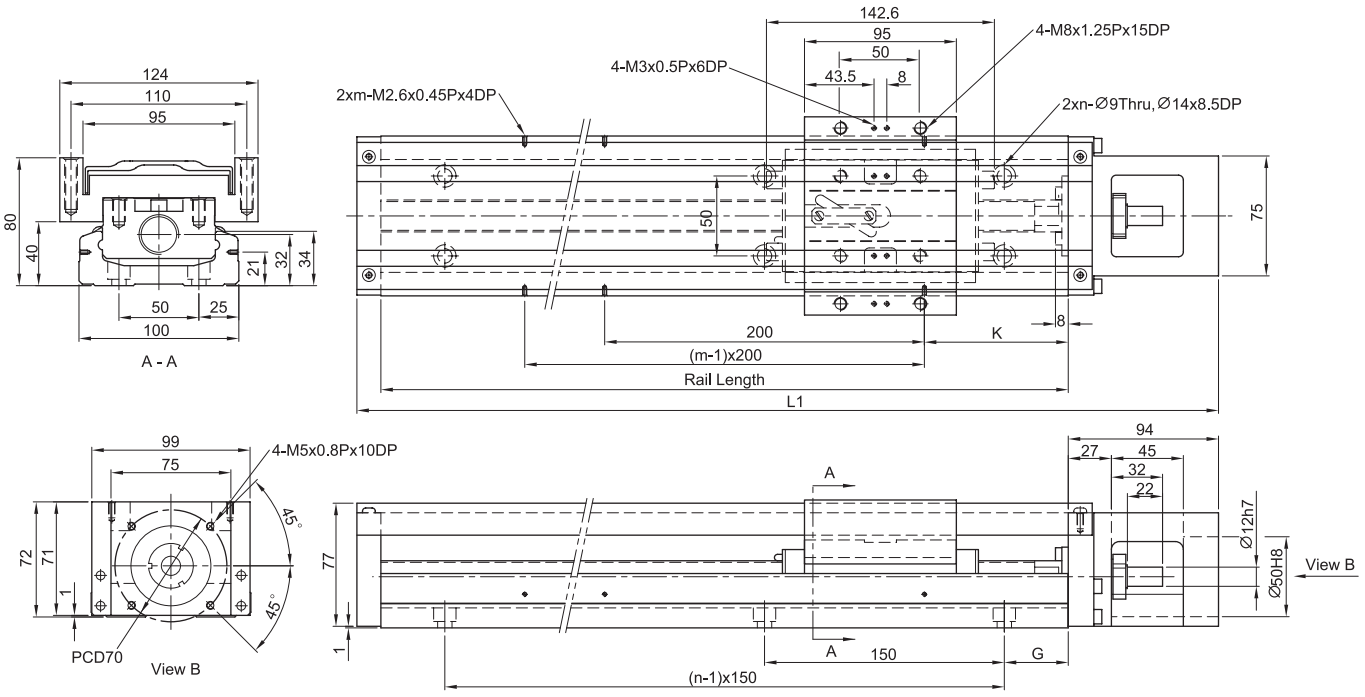

**Dimensions and Mass of the KK100 Stages without Cover**

| Rail length [mm] | Total length L1 [mm] | Maximum stroke [mm] | | G [mm] | K [mm] | n | m | Mass [kg] | |
|---|---|---|---|---|---|---|---|---|---|
| | | Block A1 | Block A2 | | | | | Block A1 | Block A2 |
| 980 | 1089 | 828 | 700 | 40 | 90 | 7 | 5 | 18,6 | 20,3 |
| 1080 | 1189 | 928 | 800 | 15 | 40 | 8 | 6 | 20,3 | 22,0 |
| 1180 | 1289 | 1028 | 900 | 65 | 90 | 8 | 6 | 22,0 | 23,7 |
| 1280 | 1389 | 1128 | 1000 | 40 | 40 | 9 | 7 | 23,6 | 25,3 |
| 1380 | 1489 | 1228 | 1100 | 15 | 90 | 10 | 7 | 25,3 | 27,0 |

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## KK100 Stages with Aluminium Cover

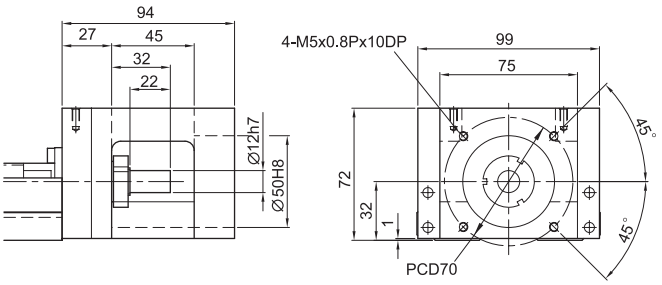



## Dimensions and Mass of KK100 Stages with Aluminium Cover

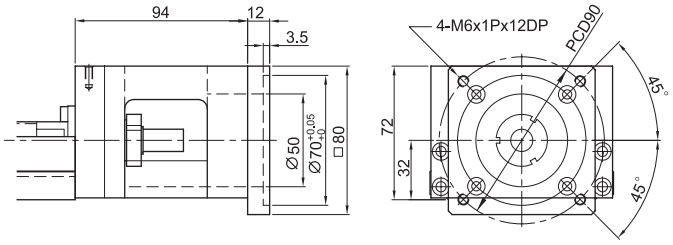
| Rail length [mm] | Total length L1 [mm] | Maximum stroke [mm] Block A1 | Block A2 | G [mm] | K [mm] | n | m | Mass [kg] Block A1 | Block A2 |
|---|---|---|---|---|---|---|---|---|---|
| 980 | 1089 | 828 | 700 | 40 | 90 | 7 | 5 | 20,4 | 22,1 |
| 1080 | 1189 | 928 | 800 | 15 | 40 | 8 | 6 | 22,2 | 23,9 |
| 1180 | 1289 | 1028 | 900 | 65 | 90 | 8 | 6 | 24,0 | 25,7 |
| 1280 | 1389 | 1128 | 1000 | 40 | 40 | 9 | 7 | 25,7 | 27,4 |
| 1380 | 1489 | 1228 | 1100 | 15 | 90 | 10 | 7 | 27,5 | 29,2 |

# Positioning Systems
## Linear Module

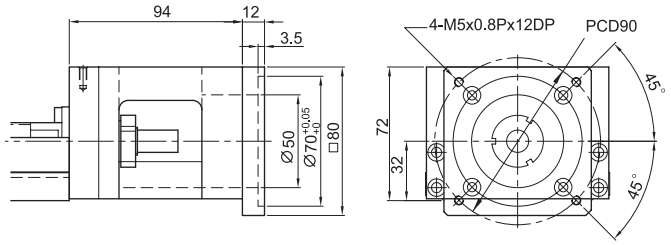
### KK100 Stages Adapter Flange F0



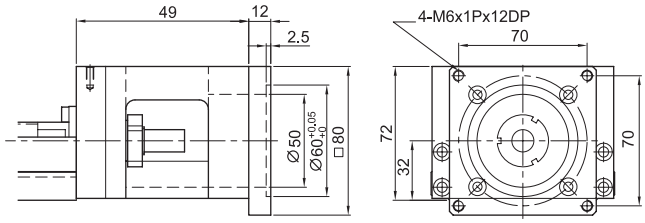

### KK100 Stages Adapter Flange F1



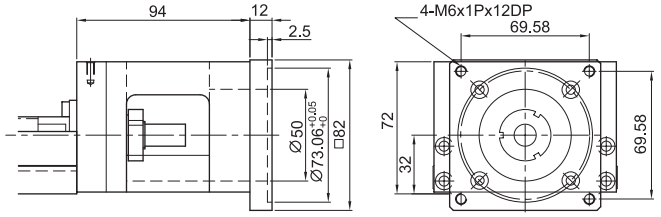

### KK100 Stages Adapter Flange F2



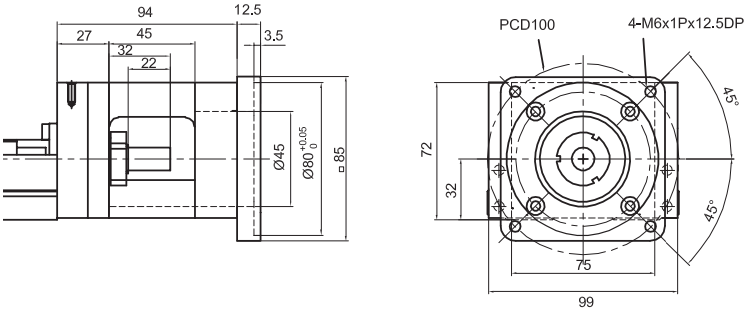

### KK100 Stages Adapter Flange F3



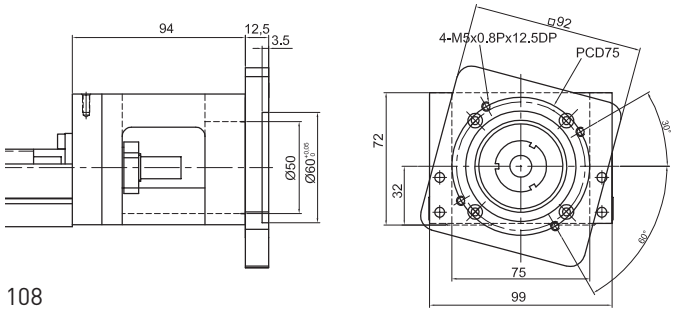

### KK100 Stages Adapter Flange F4



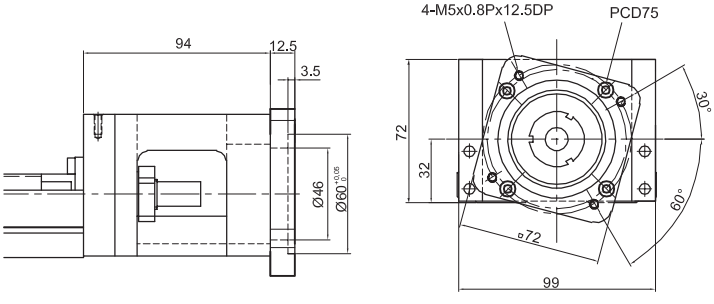

### KK100 Stages Adapter Flange F5




### KK100 Stages Adapter Flange F6




### KK100 Stages Adapter Flange F7



## 6.3 KK Linear Stage Accessories

### 6.3.1 Article Overview of Adapter Plates for KK Stages

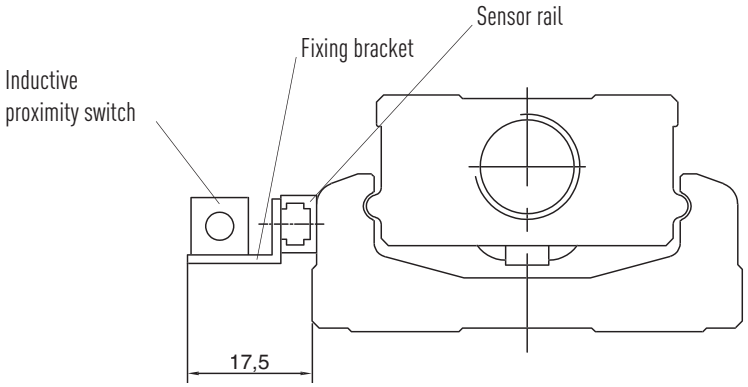
| Model | Adapter plate | Article number set (comprising adapter plate and fixing screws) |
|---|---|---|
| **KK40** | KK-40-F1 | 8-11-0205 |
| | KK-40-F2 | 8-11-0206 |
| | KK-40-F3 | 8-11-0207 |
| **KK50** | KK-50-F1 | 8-11-0209 |
| | KK-50-F2 | 8-11-0210 |
| | KK-50-F3 | 8-11-0211 |
| | KK-50-F4 | 8-11-0120 |
| | KK-50-F5 | 8-11-0212 |
| | KK-50-F6 | 8-11-0213 |
| | KK-50-F7 | 8-11-0214 |
| **KK60** | KK-60-F1 | 8-11-0215 |
| | KK-60-F2 | 8-11-0216 |
| | KK-60-F3 | 8-11-0217 |
| | KK-60-F4 | 8-11-0218 |
| | KK-60-F5 | 8-11-0219 |
| | KK-60-F6 | 8-11-0129 |
| | KK-60-F7 | 8-11-0220 |
| | KK-60-F8 | 8-11-0221 |
| | KK-60-F9 | 8-11-0222 |
| | KK-60-F10 | 8-11-0223 |
| | KK-60-F11 | 8-11-0224 |
| **KK86** | KK-86-F1 | 8-11-0225 |
| | KK-86-F2 | 8-11-0226 |
| | KK-86-F3 | 8-11-0227 |
| | KK-86-F4 | 8-11-0228 |
| | KK-86-F5 | 8-11-0229 |
| | KK-86-F6 | 8-11-0230 |
| | KK-86-F7 | 8-11-0132 |
| | KK-86-F8 | 8-11-0068 |
| | KK-86-F9 | 8-11-0231 |
| | KK-86-F10 | 8-11-0232 |
| **KK100** | KK-100-F1 | 8-11-0233 |
| | KK-100-F2 | 8-11-0234 |
| | KK-100-F3 | 8-11-0235 |
| | KK-100-F4 | 8-11-0236 |
| | KK-100-F5 | 8-11-0132 |
| | KK-100-F6 | 8-11-0237 |
| | KK-100-F7 | 8-11-0068 |

### 6.3.2 Article Overview of Sensor Rails for KK Stage

| KK sizes | Article number Sensor rail set (comprising sensor rail and fixing materials, cam switch) |
|---|---|
| **KKx4001P100A1** | 8-11-0239 |
| **KKx4001P150A1** | 8-11-0240 |
| **KKx4001P200A1** | 8-11-0241 |
| **KKx5002P150A1** | 8-11-0242 |
| **KKx5002P200A1** | 8-11-0243 |
| **KKx5002P250A1** | 8-11-0244 |
| **KKx5002P300A1** | 8-11-0245 |
| **KKx60xxP150EA1** | 8-11-0246 |
| **KKx60xxP200EA1** | 8-11-0247 |
| **KKx60xxP300EA1** | 8-11-0248 |
| **KKx60xxP400EA1** | 8-11-0249 |
| **KKx60xxP500EA1** | 8-11-0250 |
| **KKx60xxP600EA1** | 8-11-0251 |
| **KKx86xxP340A1** | 8-11-0252 |
| **KKx86xxP440A1** | 8-11-0253 |
| **KKx86xxP540A1** | 8-11-0254 |
| **KKx86xxP640A1** | 8-11-0255 |
| **KKx86xxP740A1** | 8-11-0256 |
| **KKx86xxP940A1** | 8-11-0257 |
| **KKx10020P980A1** | 8-11-0258 |
| **KKx10020P1080A1** | 8-11-0259 |
| **KKx10020P1180A1** | 8-11-0260 |
| **KKx10020P1280A1** | 8-11-0261 |
| **KKx10020P1380A1** | 8-11-0262 |

**Switch set** **8-11-0263**
consisting of fixing bracket, one inductive proximity switch and fixing materials) for use as a limit switch or reference switch
Cable length: 2m

**Switch set** **8-11-0264**
consisting of fixing bracket, one inductive proximity switch and fixing materials) for use as a limit switch or reference switch
Cable length: 4m



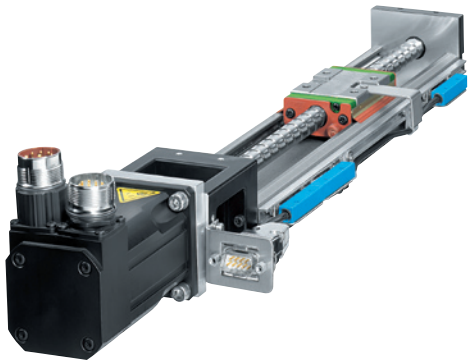
CRD DEVICES

HIWIN® Linear Technology

## 6.4 KK Stages with Motor

### 6.4.1 Scope of Delivery

Expansion of the KK stage using suitable stepping motors and servo motors with the associated amplifiers extends the KK stage into a complete positioning system. Linear stages are supplied complete with inductive limit switch, reference switch and coupling.



**Specifications for KK Stages KK40 with Stepping Motor**

| | Unit | | | |
|---|---|---|---|---|
| Rail length | mm | 100 | 150 | 200 |
| Max. stroke | mm | 36 | 86 | 136 |
| Motor | | Stepping motor (with micro-step driver) or AC servo motor | | |
| Repeatability | mm | ±0,003 | ±0,003 | ±0,003 |
| Positioning accuracy | mm | 0,02 | 0,02 | 0,02 |
| Guideway parallelism | mm | 0,01 | 0,01 | 0,01 |
| Max. speed | mm/s | 10* | 10* | 10* |
| Advance force | N | 50 | 50 | 50 |

* with stepping motor

**Specifications for KK Stages KK50 with Stepping Motor or Servo Motor**

| | Unit | | | | |
|---|---|---|---|---|---|
| Rail length | mm | 150 | 200 | 250 | 300 |
| Max. stroke | mm | 70 | 120 | 170 | 220 |
| Motor | | Stepping motor (with micro-step driver) or AC servo motor | | | |
| Repeatability | mm | ±0,003 | ±0,003 | ±0,003 | ±0,003 |
| Positioning accuracy | mm | 0,02 | 0,02 | 0,02 | 0,02 |
| Guideway parallelism | mm | 0,01 | 0,01 | 0,01 | 0,01 |
| Max. speed | mm/s | 30*/270** | 30*/270** | 30*/270** | 30*/270** |
| Advance force | N | 150 | 150 | 150 | 150 |

* with stepping motor ** with servo motor

**Specifications for KK Stages KK60, Lead 5 mm with Stepping Motor or Servo Motor**

| | Unit | | | | | | |
|---|---|---|---|---|---|---|---|
| Rail length | mm | 150 | 200 | 300 | 400 | 500 | 600 |
| Max. stroke | mm | 60 | 110 | 210 | 310 | 410 | 510 |
| Motor | | Stepping motor (with micro-step driver) or AC servo motor | | | | | |
| Repeatability | mm | ±0,003 | ±0,003 | ±0,003 | ±0,003 | ±0,003 | ±0,003 |
| Ballscrew shaft | | Dia. 12, Lead 5 | | | | | |
| Positioning accuracy | mm | 0,02 | 0,02 | 0,02 | 0,02 | 0,025 | 0,025 |
| Guideway parallelism | mm | 0,01 | 0,01 | 0,01 | 0,01 | 0,015 | 0,015 |
| Max. speed | mm/s | 75*/550** | 75*/550** | 75*/550** | 75*/550** | 75*/550** | 75*/340** |
| Advance force | N | 250 | 250 | 250 | 250 | 250 | 250 |

* with stepping motor ** with servo motor